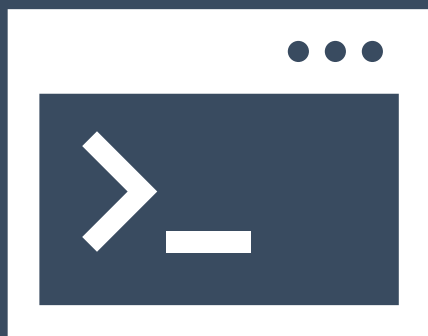


Introduction to Single-cell RNA-seq analysis

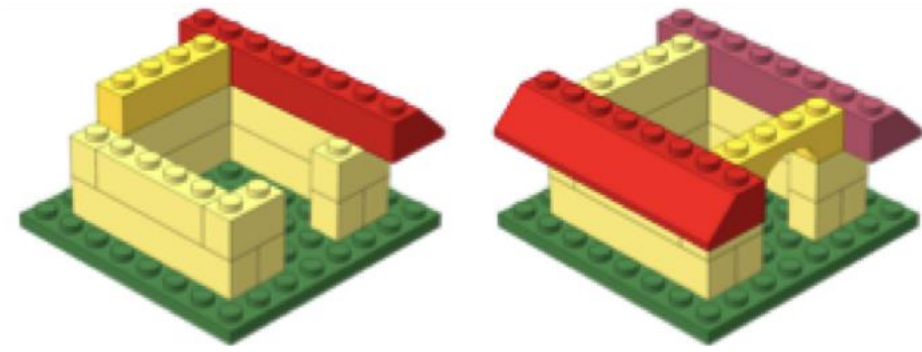
<https://tinyurl.com/hbc-intro-scrna>



Harvard Chan Bioinformatics Core



Workshop Scope



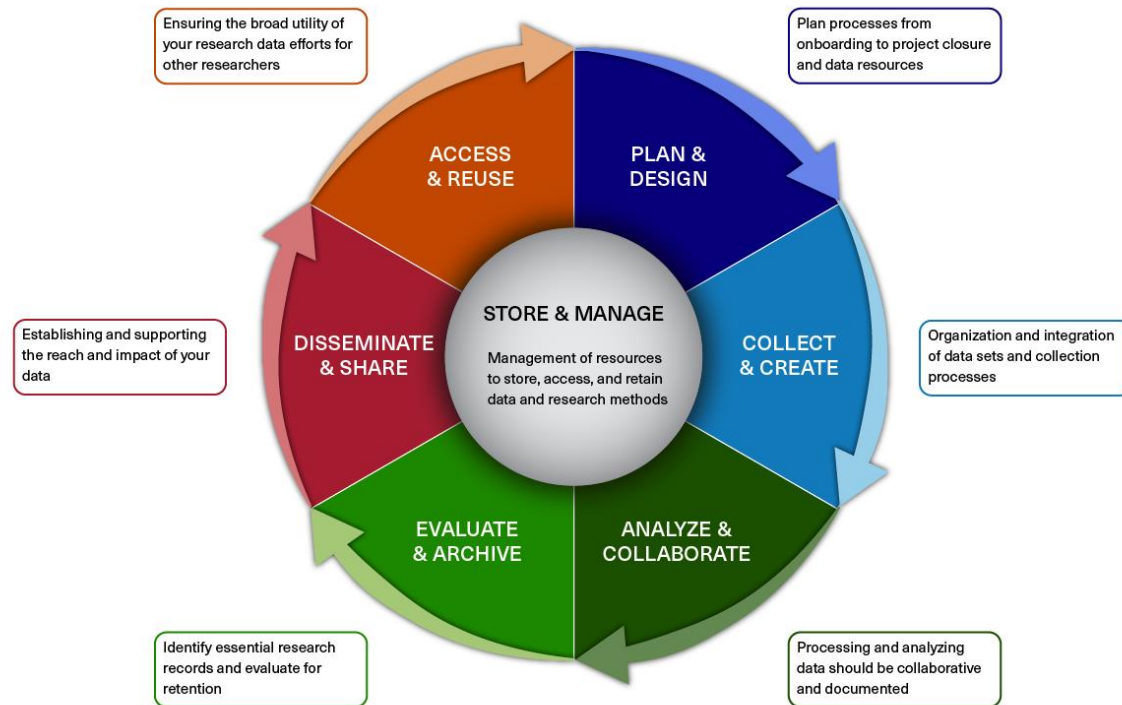
- ❖ Describe best practices for designing a single-cell RNA-seq experiment
- ❖ Describe steps in a single-cell RNA-seq analysis workflow
- ❖ Use Seurat/Scanpy and associated tools to perform analysis of single-cell expression data, including data filtering, QC, integration, clustering, and marker identification
- ❖ Understand practical considerations for performing scRNA-seq, rather than in-depth exploration of algorithm theory

Exit survey

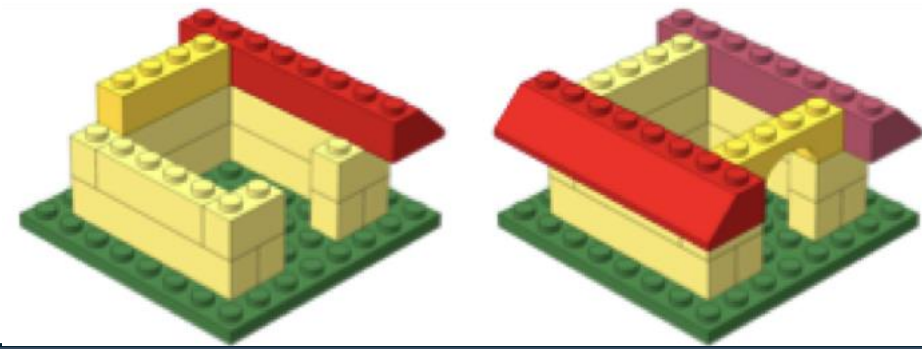
<https://tinyurl.com/scRNA-seq-online-Python>

Research Data Management (RDM)

BIOMEDICAL RESEARCH DATA LIFECYCLE



Useful resources



❖ Computational packages for single-cell analysis:

- ❖ <http://bioconductor.org/packages/devel/workflows/html/simpleSingleCell.html>
- ❖ <https://satijalab.org/Seurat/>
- ❖ <https://scanpy.readthedocs.io/>
- ❖ <https://github.com/seandavi/awesome-single-cell>

❖ Online courses:

- ❖ <https://hemberg-lab.github.io/scRNA.seq.course/>
- ❖ <https://github.com/SingleCellTranscriptomics>

❖ Resources for scRNA-seq Sample Prep:

- ❖ <https://www.protocols.io/>
- ❖ <https://support.10xgenomics.com/single-cell-gene-expression/sample-prep>
- ❖ <https://community.10xgenomics.com/>

Better RDM practice benefits you

❖ HMS Data Management LMA

❖ **Webpage:** <https://datamanagement.hms.harvard.edu>

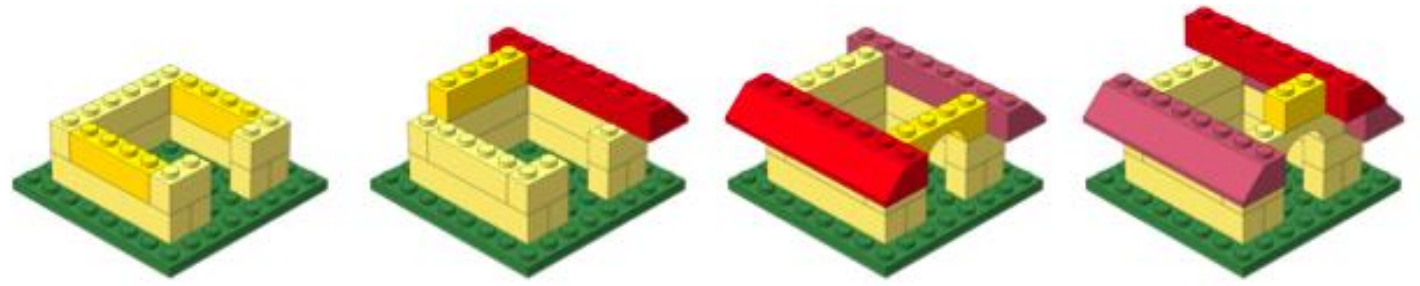
❖ Sign up for quarterly email updates

❖ Harvard-wide Research data Management

❖ <https://researchdatamanagement.harvard.edu/>

Date	Time	Event	Location
Aug 27	11am	Digital Preservation and Research Data	Zoom
Aug 28	10am	Introduction to GIS	Lamont Library - Media Lab Annex
Sep 2	10am	Introduction to GIS	Lamont Library - Media Lab Annex
Sep 3	10am	LMA Research Data Management Working Group Monthly Meeting	Zoom
Sep 11	9am	HKS Data Carpentry Day 1: Tidy Data, OpenRefine, and Intro to R	HKS Wexner Building
Sep 14	2pm	protocols.io Webinar: Tips and Tricks	Zoom
Sep 16	12pm	Data Discussions: Let's Talk Data Services	Zoom
Sep 22	12pm	Biomedical and Public Health Research Data Management: Concepts, Tools, and Strategies	Countway Library, Classroom L1-028

Keep building!



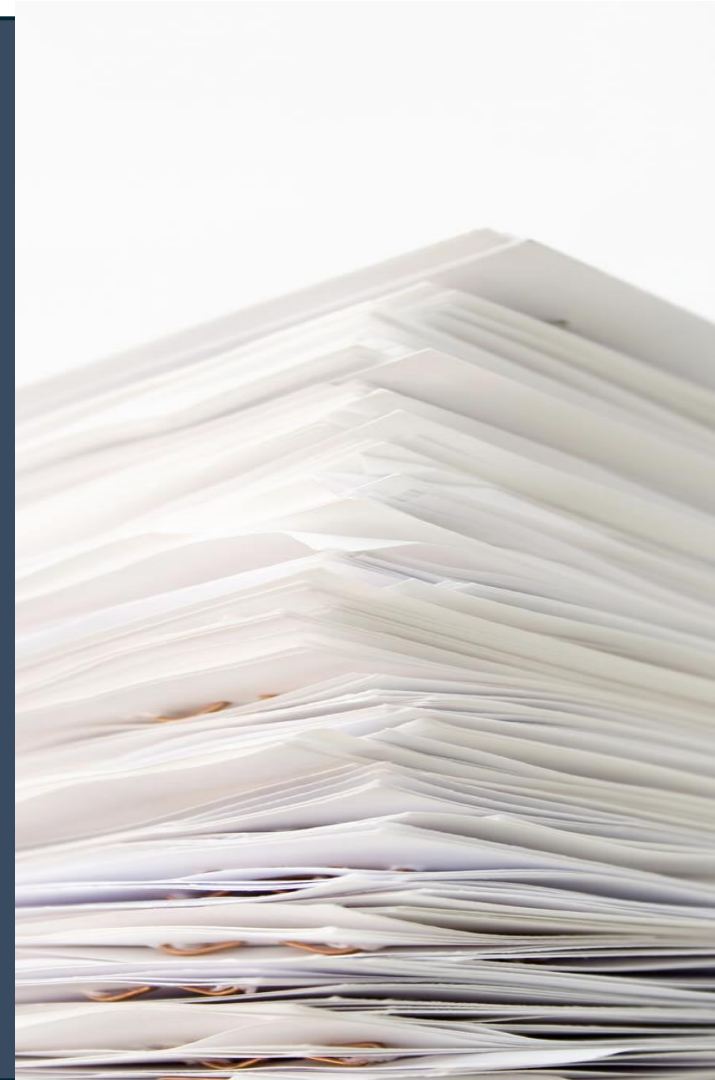
Workshop	Dates	Time	Location	Registration Page*
Introduction to R	August 28, September 1, 4 & 8 2026	10am-12pm	Zoom	Register here
Introduction to Python	November 10, 13, 17 & 20 2026	10am-12pm	Zoom	Register here

Workshop	Pre-requisite*	Dates	Time	Location	Registration Page**
Introduction to Spatial Transcriptomics	Introduction to R	October 6, 9, 13 & 16, 2026	9:30am-12pm	Zoom	Register here
Tools for Reproducible Research	Introduction to R	October 27, 30 & November 3, 2026	9:30am-12pm	Zoom	Register here

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Talk to us early!

Involvement in study design to optimize experiments



More Information

- ❖ *HBC training materials: <https://hbctraining.github.io/main>*
- ❖ *HBC website: <https://hsph.harvard.edu/research/bioinformatics/>*

Contact Us

Sign up for our mailing list:

<https://tinyurl.com/hbc-training-mailing-list>

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu